MECH5200 Numerical Methods for PDEs : *Video 11: 1D Finite Difference Mappings – Theory and MATLAB*

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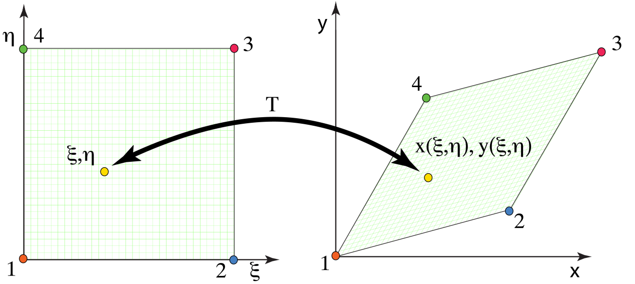
Idea

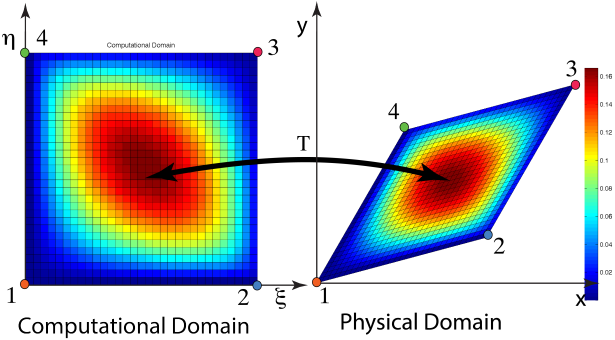
- So far, we've only considered **square** and **simple 1-D unit** domains.
  - domain is something other than a square/rectangle/unit line?
  - finite difference is not aligned with $x-$ and $y-$ directions?
  - nodes are along a curved line?
- So far, the nodes have been *equally spaced*.
  - What about refining locally to get better answers?
  - How to we setup FD with varying node refinement?



Idea

Define a **mapping**, T, between a **computational** and **physical** domain.

- Mathematical or numerical (computational $\rightarrow$ physical)
- Transform from the physical $\rightarrow$ computational.
- Solve the PDE on the computational domain
- Map the computational results to the physical domain.

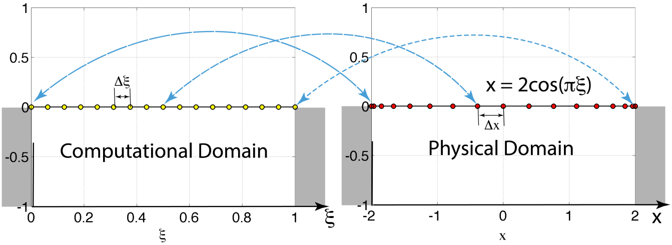


1-D Mapping

- Let's use the 1-D string example:
  - Let's define the **physical** and **computational** domains
  - **Computational:** Define computational domain: $\xi = (0, 1)$.
  - Define the physical space (non-uniform node spacing in this case) as follows:

$$x(\xi) = -2cos(\pi\xi)$$

  - The 'real' string is 4 units long.
  - The 'real' string is refined at sides.



1-D Mapping

- We'd like to solve:
$$\nabla^2 u(x(\xi)) = \frac{\partial^2 u}{\partial x^2} = f$$
using the evenly spaced discretization of the **computational** domain rather than the non-uniform discretization of the physical domain.
- To do this, we need to recognize that:
$$u(x) = u(x(\xi))$$
- **Goal**: what is the *equivalent* equation to solve on the computational domain that will yield the correct answer at the nodes on the physical domain. ie. What equation can we solve in ξ domain and get the right answer at each node?

1-D Mapping

- Let's start by expressing $\frac{\partial(\cdot)}{\partial x}$ in the computational domain.
- The goal is to find the derivatives w.r.t. ξ rather than x. This way, we can perform finite differences w.r.t. ξ.
- Taking the first derivative:

$$\frac{\partial(\cdot)}{\partial x} = \frac{\partial(\cdot)}{\partial \xi}\frac{d\xi}{dx}$$

Similarly, we can also write a similar expansion:

$$\frac{\partial(\cdot)}{\partial \xi} = \frac{\partial(\cdot)}{\partial x}\frac{dx}{d\xi}$$

1-D Mapping

- So, the first derivative is:

$$\frac{\partial(u)}{\partial x} = \frac{\partial(u)}{\partial \xi}\frac{d\xi}{dx}$$

- Now, let's try the second derivative:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial\left(\frac{\partial(u)}{\partial x}\right)}{\partial x} = \frac{\partial\left(\frac{\partial(u)}{\partial x}\right)}{\partial \xi}\frac{d\xi}{dx} = f$$

- Substituting in the first derivative result from above:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}\cdot\frac{d\xi}{dx}\right)\frac{d\xi}{dx} = \frac{d\xi}{dx}\frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}\cdot\frac{d\xi}{dx}\right) = f$$

1-D Mapping

- Now, we need to apply the chain rule to the “red” terms:

$$\frac{\partial^2 u}{\partial x^2} = \frac{d\xi}{dx}\frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}\right)\cdot\frac{d\xi}{dx} + \frac{d\xi}{dx}\frac{\partial u}{\partial \xi}\cdot\frac{\partial}{\partial \xi}\left(\frac{d\xi}{dx}\right) = f$$

- Simplify terms a little (collect like terms, re-color to collect similar terms):

$$\frac{\partial^2 u}{\partial x^2} = \left(\frac{d\xi}{dx}\right)^2\frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}\right) + \frac{d\xi}{dx}\frac{\partial u}{\partial \xi}\cdot\frac{\partial}{\partial \xi}\left(\frac{1}{\frac{dx}{d\xi}}\right) = f$$

- Continuing:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{\left(\frac{dx}{d\xi}\right)^2}\frac{\partial^2 u}{\partial \xi^2} - \frac{1}{\left(\frac{dx}{d\xi}\right)^3}\frac{\partial^2 x}{\partial \xi^2}\frac{\partial u}{\partial \xi} = f$$

1-D Mapping

- Using J to represent the Jacobian of the mapping, $J = \frac{dx}{d\xi}$, the equation is :

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{J^2}\frac{\partial^2 u}{\partial \xi^2} - \frac{1}{J^3}\frac{\partial^2 x}{\partial \xi^2}\frac{\partial u}{\partial \xi} = f$$

- The result is an **equivalent** equation in the computational domain as that which must be satisfied in the physical domain.
- The components in blue are those corresponding to the geometry transformation. Those in red correspond to the derivative of the solution taken in the computational domain.

1-D Mapping

- To implement the equation in a finite difference solution method in the **computational** domain:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{J^2}\frac{\partial^2 u}{\partial \xi^2} - \frac{1}{J^3}\frac{\partial^2 x}{\partial \xi^2}\frac{\partial u}{\partial \xi} = f$$

- Substitute the second order, central difference equations for the first and second derivatives:

$$\left(\frac{\partial^2 u}{\partial x^2}\right) = \frac{1}{J^2}\left(\frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta \xi^2}\right) - \frac{1}{J^3}\frac{\partial^2 x}{\partial \xi^2}\left(\frac{u_{i+1} - u_{i-1}}{2\Delta x}\right) = f_i$$

- Now, all that's left is to construct the matrices.

1-D Mapping

Let's generalize for a given row i of the matrix:

$$\left(\frac{\partial^2 u}{\partial x^2}\right) = \frac{1}{J^2}\left(\frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta\xi^2}\right) - \frac{1}{J^3}\frac{\partial^2 x}{\partial\xi^2}\left(\frac{u_{i+1} - u_{i-1}}{2\Delta x}\right) = f_i = 1$$

The A-Matrix entries are:

$$\begin{aligned}
A(row_i, column_i) &= -2 \cdot \frac{1}{J^2}\left(\frac{1}{\Delta\xi^2}\right) \\
A(row_i, column_i + 1) &= 1 \cdot \frac{1}{J^2}\left(\frac{1}{\Delta\xi^2}\right) - \frac{1}{J^3} \cdot \frac{\partial^2 x}{\partial\xi^2} \cdot \left(\frac{1}{2\Delta\xi}\right) \\
A(row_i, column_i - 1) &= 1 \cdot \frac{1}{J^2}\left(\frac{1}{\Delta\xi^2}\right) + \frac{1}{J^3} \cdot \frac{\partial^2 x}{\partial\xi^2} \cdot \left(\frac{1}{2\Delta\xi}\right) \\
f(row_i) &= 1
\end{aligned}$$

1-D Mapping

- For our example:

$$\begin{aligned} \nabla^2 u(x(\xi)) &= f \\ x &= -2cos(\pi\xi) \end{aligned}$$

- The mapping coefficients can be **analytically computed**:

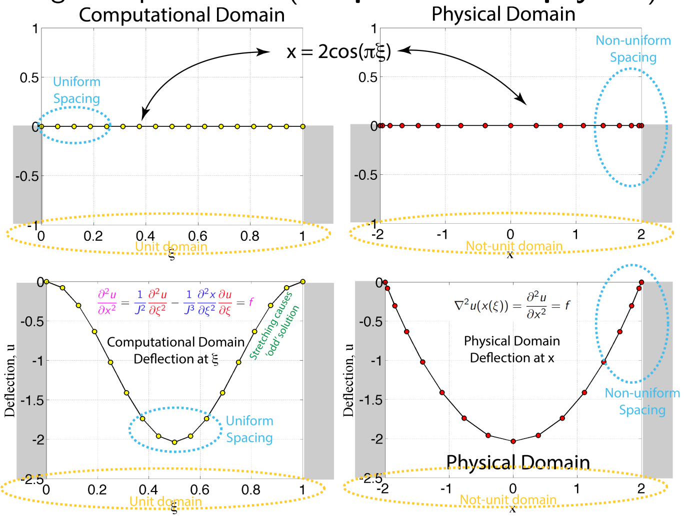
$$\begin{aligned} \frac{dx}{d\xi} &= 2\pi \sin(\pi\xi) \\ \frac{d^2x}{d\xi^2} &= 2\pi^2 \cos(\pi\xi) \end{aligned}$$

- **Numerical Mapping:** Alternatively, if we only had the points in the physical domain (ie. not provided with an analytical mapping), we could simply use finite difference formulae to calculate the derivatives, e.g.:

$$\frac{dx}{d\xi} = \frac{x_{i+1} - x_{i-1}}{2\Delta\xi}$$

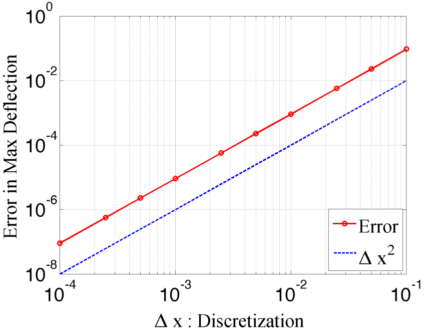
1-D Mapping

- 1-D string example solution (**computational** vs. **physical**):



1-D Mapping

- Let's check our infinity-norm convergence (max error) :



- The convergence is second order (i.e., the solution converges as expected)

What have we learned?

- We can transform an equation from a tricky physical domain to a simpler reference computational domain.
- We need to develop a new governing equation for the computational domain that **represents** the physical domain equation but on a uniform grid/spacing.
- Once we have the mapping & new equation, the solution to the finite difference problem is identical to what was done before.
- The mapping can be computed numerically or analytically (depending on complexity).